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Key Points:

- SamudrACE is an AI emulator of the GFDL CM4 coupled global climate model (GCM), trained on 150 years of pre-industrial output
- Like traditional coupled GCMs, SamudrACE's components were developed independently by distinct teams and coupled via fine-tuning
- Running on a single NVIDIA H100 GPU, SamudrACE can simulate 1,500 years per day

Supporting Information:

Supporting Information may be found in the online version of this article.

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SamudrACE: Fast and Accurate Coupled Climate Modeling With 3D Ocean and Atmosphere Emulators

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Abstract Traditional numerical global climate models simulate the full Earth system by exchanging boundary conditions between separate simulators of the atmosphere, ocean, sea ice, land surface, and other geophysical processes. This paradigm allows for distributed development of individual components within a common framework, unified by a coupler that handles translation between realms via spatial or temporal alignment and flux exchange. Following a similar approach adapted for machine learning-based emulators, we present SamudrACE: a coupled global climate model emulator which produces centuries-long simulations at 1-degree horizontal, 6-hourly atmospheric, and 5-daily oceanic resolution, with 145 2D fields spanning 8 atmospheric and 19 oceanic vertical levels, plus sea ice, surface, and top-of-atmosphere variables. SamudrACE is highly stable and has low climate biases comparable to those of its components with prescribed boundary forcing, with realistic variability in coupled climate phenomena such as ENSO that is not possible to simulate in uncoupled mode.

Plain Language Summary Climate scientists use computer models to understand how different parts of the Earth system, like the atmosphere and ocean, work together. Traditionally, these models are built with separate components that exchange information. We applied this same approach to new, much faster models based on artificial intelligence (AI). We connected an AI atmosphere model (called ACE) to an AI ocean model (called Samudra) to create a new, coupled AI model called SamudrACE, capable of simulating the full Earth climate evolution. Our combined model runs stably for centuries, producing accurate, high-quality climate simulations. A key success is that by linking the ocean and atmosphere, SamudrACE can realistically simulate complex, large-scale climate phenomena like the El Niño-Southern Oscillation (ENSO), something the individual AI models cannot do on their own. This work demonstrates a successful strategy for building fast and powerful AI-based tools to study long-term climate evolution more efficiently.

1. Introduction

The advent and success of machine learning (ML)-based weather prediction (Bi et al., 2023; Lam et al., 2023; Pathak et al., 2022) has led to similarly data-driven global atmosphere emulators trained on output of numerical models, such as the atmosphere-only version of the Ai2 Climate Emulator (ACE) (Watt-Meyer et al., 2023). Since then, atmosphere model emulators have continued to mature and support Atmosphere Model Intercomparison Project (AMIP) (Gates, 1992) compatible simulations (Chapman et al., 2025; Kochkov et al., 2024; Watt-Meyer et al., 2025).

We demonstrate early progress toward a global climate model (GCM) emulator capable of running the Coupled Model Intercomparison Program (CMIP) DECK simulation suite (Eyring et al., 2016) by coupling modular atmosphere, sea ice, land, and ocean emulators. This could later be extended to incorporate other components of the Earth system (e.g., biogeochemical processes).

Coupled atmosphere and ocean emulation is needed to learn and generate realistic climate trends (e.g., through the time-evolving spatial patterns of ocean heat uptake and sea-surface temperature rise). It is also needed to generate the variability in physical phenomena that emerge through the realistic interaction and coupled evolution of

atmospheric surface forcing and upper ocean response, such as El Niño–Southern Oscillation (ENSO) variability (Zebiak & Cane, 1987).

Several recent papers have incorporated simplified forms of ocean coupling into ML atmospheric emulators, for example, by use of a physically-based slab ocean model (Clark et al., 2025) expressed in PyTorch (Paszke et al., 2019), or by prognostically emulating sea surface temperature (SST) (Cresswell-Clay et al., 2025), or with the addition of near-surface temperature on a limited number of upper-ocean levels (C. Wang et al., 2024). While these approaches enable accurate seasonal forecasts (C. Wang et al., 2024) and stable simulation of present-day (Cresswell-Clay et al., 2025) and CO₂-modulated (Clark et al., 2025) equilibrium climate, their simplified ocean representations are insufficient to support accurate coupled atmosphere–ocean variability such as ENSO.

To go further toward emulation of climate-coupled phenomena requires resolving the full extent of the ocean in order to simulate ocean circulation and response to atmospheric forcing on annual and decadal timescales. Fortunately, three-dimensional ML ocean emulators have been recently developed for data-driven ocean forecasting on timescales up to 1–2 years (Chen et al., 2023; El Aouni et al., 2024; Xiong et al., 2023; X. Wang et al., 2024) and for longer-running simulations forced by specified time-evolving atmospheric conditions (Dheeshjith et al., 2025). Specifically, Samudra (Dheeshjith et al., 2025) stably emulates GFDL's Ocean Model v4 (OM4) and reproduces ocean dynamics on decadal timescales when forced from above by the net downward heat flux (at the ocean surface, or beneath sea ice where present) and surface wind stresses. Similarly, successful emulators have been developed for sea ice (Durand et al., 2024).

However, these advances in component model emulation do not necessarily enable their coupling. So far, there has not been a data-driven approach capable of successful 3D coupled emulation of the full vertical extents of the atmosphere and ocean. Full Earth system emulation introduces complexities beyond component modeling, including reconciling distinct vertical coordinates, managing new internal boundaries, and selecting between physical or latent space coupling. Crucially, the system must bridge vast spatiotemporal scales—from fast weather to slow ocean dynamics—to capture emergent coupled phenomena like ENSO.

In this paper, we present SamudrACE, which is constructed by coupling the ACE2 3D atmosphere emulator to the Samudra 3D ocean emulator which has been extended to predict sea ice concentration and thickness. Both components are emulated at 1° lat/lon horizontal resolution. We train SamudrACE to emulate a fully coupled 200-year simulation (For convenience in the remainder of the text and figures, we use “CM4” interchangeably to denote both the coupled GCM and the 200-year simulation output from that model.) by the GFDL CM4 physics-based coupled GCM with constant pre-industrial greenhouse gas and aerosol concentrations and a repeating annual cycle of insolation (Held et al., 2019).

2. Materials and Methods

2.1. Component Emulators: Samudra and ACE2

Analogous to CM4's coupling (Balaji, 2011) of the components AM4/LM4 (Zhao et al., 2018) to OM4/SIS2 (Adcroft et al., 2019), SamudrACE is composed of two independent component emulators and a coupler that handles communication between the two. Specifically, SamudrACE couples the ACE2 (Watt-Meyer et al., 2025) atmosphere and land surface emulator to the Samudra (Dheeshjith et al., 2025) ocean emulator. To facilitate this coupling, both ACE2 and Samudra are pretrained from random weight initializations on the CM4 simulation outputs with prescribed forcings, a step that we call “uncoupled pretraining” and describe in detail below. Additionally, we modified Samudra to prognose sea ice concentration and mass. After pretraining, ACE2 and Samudra are coupled and fine-tuned in two stages: first only updating Samudra's weights and then jointly updating both components. SamudrACE is the end result of this uncoupled pretraining and coupled fine-tuning pipeline. Tables S1 and S2 in Supporting Information S1 provide a complete listing of the SamudrACE component outputs.

2.2. The SamudrACE Coupler

Figure 1 provides a schematic of the SamudrACE coupler. The ACE2 model simulates the atmosphere in 6-hr increments. At the end of each step, prognostic variables, such as temperature, are saved as an instantaneous snapshot. In contrast, diagnostic boundary fluxes are calculated as the average value over that same 6-hr interval. For each forward step, ACE2 is forced by the sea ice fraction and SST from time t . Once ACE2 completes 20

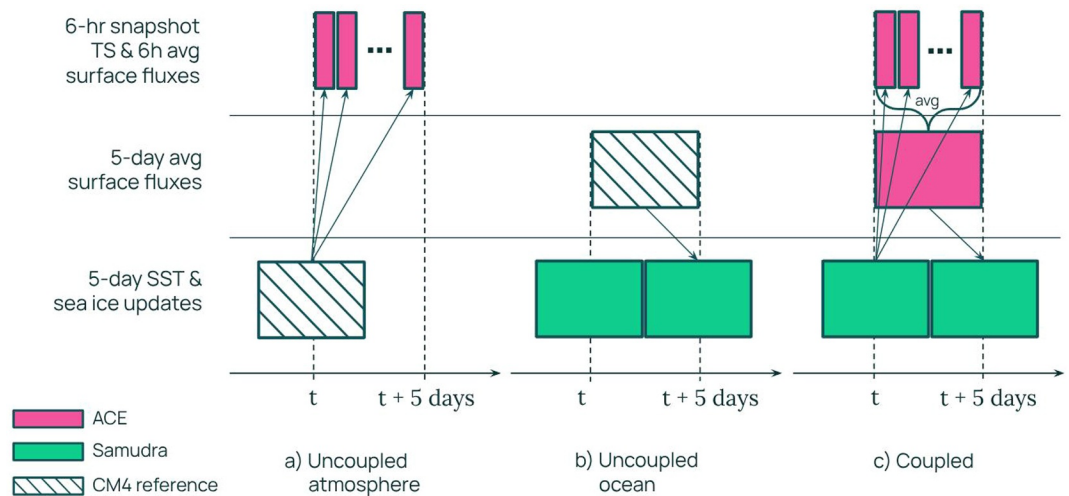


Figure 1. A single 5-day forward step in uncoupled (a, b) and coupled (c) modes. In uncoupled mode, ACE and Samudra are forced by the appropriate CM4 reference fields. Uncoupled ACE is forced by the reference 5-day average SST and sea ice concentration, stepping forward 6 hr at a time until reaching 5 days, at which point the next 5-day average forcing is prescribed. For uncoupled Samudra, we use the 5-day average of the CM4 reference wind stress, precipitation, and surface fluxes as prescribed forcing inputs. In coupled mode, aggregation of the diagnostic 6-hr average surface boundary conditions to 5-day averages is done online as ACE completes 20 forward steps. The generated 5-day average is then passed as input to Samudra, which generates a single 5-day forward step. Samudra's generated SST and sea ice states will then be used to force ACE in the next iteration of the coupler loop.

forward steps the coupler aggregates the 6-hr average boundary fluxes into a single 5 days average. The generated ocean surface fluxes of energy, moisture and momentum are then used to force a single step of Samudra, which evolves the ocean state forward in time by 5 days to catch up to the atmosphere. The new SST state is then prescribed onto the final generated global surface temperature T_S state in preparation to resume atmosphere forward stepping.

The 6-hr atmosphere and 5-day ocean timesteps reflect their respective timescales of change on the $O(100 \text{ km})$ grid. While this 20-fold timestep difference complicates training, it also enables realistic variability in both components. The SamudrACE coupler is heavily influenced by GFDL's Flexible Modeling System (FMS) coupler (Balaji, 2011) used in CM4 simulations. However, unlike in typical physics-based coupled models, ACE2 internally predicts the surface flux components, rather than these being computed by the ocean or land surface models.

Our choice of coupling procedure is an important physics-informed design element of SamudrACE. The flow of information at component interfaces is column-local and constrained so that the atmosphere forces the ocean with surface fluxes and the ocean forces the atmosphere with SST and sea ice. Enforcing this exchange of information between components in physical state space via a coupling procedure avoids the exchange of nonphysical latent representations between the two models. ACE2 and Samudra were each designed from the start with coupling in mind by learning the diagnostic variables needed for a physically justifiable coupling procedure and temperature and salinity budget closure.

2.3. Pretraining ACE2

We follow ACE2's training protocol (Watt-Meyer et al., 2025) with two additional diagnostic variables—surface zonal and meridional wind stress. Training data come from the atmosphere component output of CM4 at 6-hourly temporal resolution, with the exception of surface temperature over ocean, sea ice fraction f_i , and ocean fraction f_o (derived from f_i). For each 5-day window aligned with the ocean data's timesteps, the ocean-covered grid cells of 6-hourly T_S are held fixed at their values at the beginning of the 5-day window, while allowing grid cells with sea ice or land to evolve with the usual 6-hr timestep. Similarly, all 20 snapshots of f_i and f_o will be identical to the first snapshot. We provide ACE2 pretraining details in Text S4 in Supporting Information S1 and training losses and inference RMSE in Figure S1b in Supporting Information S1.

2.4. Pretraining SamudraI

Our implementation of Samudra (Dheeshjith et al., 2025) modifies the original protocol as follows:

1. Pretraining data set: We pretrain Samudra on 5-day mean OM4 output. For consistency, this is taken from the same segment of our reference CM4 simulation as used for ACE2 pretraining.
2. Prognostic Variables and Forcing: We expanded Samudra to include sea ice concentration (SIC) and thickness, which reduced Arctic upper-layer salinity biases (Figure S21 in Supporting Information S1). We refer to this version as “SamudraI.” The model is forced by ACE surface conditions (wind stress, heat, and water fluxes) above the sea ice, time-averaged over 5-day blocks. The original Samudra was forced only by the top-of-ocean heat flux and surface stress, calculated below any sea ice.
3. Time Stepping: We simplified the architecture to use a single input timestep ($P = 1$) and four autoregressive steps ($N = 4$) to reduce computational cost without degrading global time-mean bias.

To achieve a fully coupled global climate emulator, we must also predict sea ice. We treat sea ice as the interface between the ocean's slow time step and the atmosphere's fast time step. SamudraI updates the sea ice concentration state every 5 days, in turn determining where ACE2 is allowed to prognose surface temperature over sea ice for the subsequent 5-day window. We use sea ice concentration—defined as the sea ice grid cell fraction divided by (1—land grid cell fraction)—in order to avoid over-representing grid points with minimal ocean cover, and find that doing so reduces biases near coastlines where ocean fraction is small. SamudraI was trained using the same constant learning rate schedule as ACE2 (Text S4 in Supporting Information S1). We show SamudraI pretraining loss and inference RMSE in Figure S1a in Supporting Information S1.

2.5. Fine-Tuning SamudrACE

Once ACE2 and SamudraI pretraining is complete, we select the checkpoints with the lowest normalized channel-mean root mean square error (RMSE) over all epochs for each respective model. Together, the coupled emulators have a combined total of nearly 600 million parameters. Prior to fine-tuning, initial coupled surface temperature biases are larger than the biases observed in either uncoupled model, especially in the ENSO region and above sea ice (Figure S16a in Supporting Information S1). Text S1 in Supporting Information S1 discusses this bias and its sensitivity to specific data processing choices which impact the consistency between uncoupled pretraining and coupled fine-tuning.

Coupled fine-tuning proceeds in two stages. In the first stage, we fine-tune SamudraI by optimizing the loss of four forward steps (20 days) while coupled to the pretrained ACE2 model. Throughout this first phase of coupled fine-tuning, ACE2's model weights are held fixed and atmospheric fields do not contribute to the training or validation loss. However, atmosphere fields do contribute to the “best” checkpoint selection (Figure S1c in Supporting Information S1), which is the SamudraI checkpoint leading to the lowest normalized channel-mean RMSE across all ocean and atmosphere fields. Coupled surface temperature biases are greatly reduced after fine-tuning Samudra in coupled mode (Figure S16b in Supporting Information S1), with a reduction in global time mean RMSE from 2.265 to 0.355 K. Much of this improvement is realized after only a single epoch of fine-tuning.

In the second stage, we continue fine-tuning with the best SamudraI checkpoint found in stage one. Since ACE2 was not updated during the first phase, it is initialized from its best pretraining checkpoint. In contrast to the first stage, weights for both ACE2 and SamudraI are jointly optimized with a cosine-annealing decay schedule (Loshchilov & Hutter, 2016). The simulation trajectory again spans four 5-day ocean steps, during which the atmosphere advances through a sequence of eighty 6-hr steps. As before, we optimize the loss of all four ocean steps, but now we add the loss of the first two of the 80 atmospheric steps (Figure S1d in Supporting Information S1). See Text S4 in Supporting Information S1 for additional details.

2.6. Data Sets

Our training and evaluation data sets are from a 200-year preindustrial control simulation from GFDL's Climate Model v4 (CM4) (Held et al., 2019), rerun to save high frequency data, starting from year 151 of the original CMIP6 simulation. Atmosphere and ocean data sets were conservatively remapped to a 1° Gaussian grid with 8 terrain-following atmospheric layers and 19 ocean layers with constant-depth interfaces for emulator training and testing. We use the first 155 years of output for training, the next 5 years for validation, and the remaining 40 years of data are held out for testing. This training duration is supported by sensitivity analysis (Text S3 and Figures S17

and S18 in Supporting Information S1) showing that longer records are essential for capturing internal variability like ENSO, even as time-mean error metrics stabilize with shorter data sets.

The AM4 atmospheric fields and SIS2 sea ice concentration were output fully consistently with ACE, with instantaneous snapshots of prognostic variables output every 6 hours. All surface and top-of-atmosphere fluxes were output as six-hour time-averages between these snapshots. This enables the surface fluxes to be accumulated over 20 atmospheric steps into 5-day averages suitable for forcing the ocean emulator.

The OM4 ocean fields were all output as 5-day averages, rather than instantaneous snapshots. This includes the sea-surface temperature, used to force the ACE atmosphere model. For the purposes of this paper, these 5-day averages are used as an estimate of the ocean state at the midpoint of the averaging interval.

Achieving consistent land, ice, and open ocean masking between the emulators is a crucial step. Just as a traditional coupled GCM must reconcile differences in horizontal resolution when coupling the ocean to the atmosphere, emulators must do so either during data preparation or via an online procedure. In this work, we emulate CM4 with a common 1° horizontal resolution for both the atmosphere and ocean and therefore handle mask alignment during the data preparation step. Early versions of the coupled emulator suffered from relatively large surface temperature biases on grid cells with coastal sea ice due to small inconsistencies in the sea ice from the separate atmosphere and ocean data processing pipelines which resulted in the use of inconsistent sea ice during uncoupled pretraining of Samudra and ACE. These errors were corrected by instead pretraining Samudra with the sea ice resulting from the atmosphere data processing pipeline. We note that coupled fine-tuning of ACE was also an effective way to reduce these biases, allowing ACE to adapt to Samudra's sea ice representation in coupled mode. In Text S1 in Supporting Information S1 we provide additional discussion of our data processing choices and their effects on the coupled emulator.

3. Results

SamudrACE can stably emulate CM4 with low bias in the time series of global means and time mean spatial error patterns over a held-out 40-year inference period. These biases are of similar magnitude to the biases of the uncoupled component emulators. Using a single NVIDIA H100 GPU, a SamudrACE simulation generates around 1,500 SYPD (6.4 days per second), a 3,750× decrease in energy usage when compared to the CM4 simulation at 16 SYPD (0.068 days per second) using 6,399 CPU cores on AMD EPYC 7H12 processors. Beyond the brief summary provided in Sections 3.1–3.2, we provide a comprehensive accounting of bias and other details in Supporting Information S1.

3.1. Climate Mean State

Like its component ML emulators, SamudrACE skillfully emulates climate mean states. Its overall time mean state is accurate in terms of the RMSE of the generated time mean (Figure S2 in Supporting Information S1) and globally averaged annual mean series (Figures S3 and S4 in Supporting Information S1).

SamudrACE's atmosphere component faithfully reproduces CM4's time-mean precipitation and surface temperature over the 40-year held-out test period (Figure 2). Time mean biases are small for both precipitation and surface temperature around the globe, with absolute maxima of just over 2 mm/day or 3 K, respectively. Precipitation biases are concentrated in the tropics while surface temperature biases are pronounced in areas of sea ice and topographic features. SamudrACE also accurately captures the distribution of daily precipitation (Figure S15 in Supporting Information S1), although it slightly underestimates extreme precipitation, a typical flaw in emulators that are trained to minimize the mean squared error loss (Perkins et al., 2025).

The ocean component similarly shows low time-mean spatial bias patterns, such as in the topmost layer currents and in the vertical structure of near-surface zonal sea water currents (Figures S5 and S6 in Supporting Information S1). Figure S22 in Supporting Information S1 provides additional time-mean bias maps for sea surface height, column-integrated ocean heat content, and column-integrated ocean salt content.

The Atlantic Meridional Overturning Circulation (AMOC) plays a vital role in regulating the global climate as a critical pathway for the flux of warm water into the North Atlantic. SamudrACE emulates the vertical structure of CM4's AMOC with low bias in the 200-year time mean state (Figure S12a in Supporting Information S1) and corresponding time series of AMOC anomalies in uncoupled SamudraI, SamudrACE and CM4 (Figure S12b in

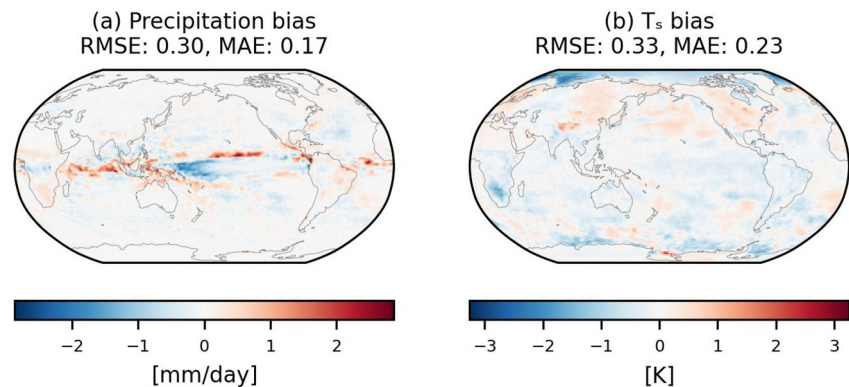


Figure 2. Spatial bias patterns of the generated 40-year time-mean (a) precipitation and (b) surface temperature, computed as the difference between the time mean of the emulated output and the time mean of the reference simulation over the held-out inference period.

Supporting Information S1). Whereas uncoupled SamudraI lacks sufficient variability in the AMOC anomaly series, SamudrACE shows similar low frequency variability to CM4.

While neither of SamudrACE's components guarantee exact closure of global energy budgets, they nonetheless generate budgets that are consistent with the reference simulation. Global ocean heat content in SamudrACE is stable over time and does not drift in a way that is inconsistent with CM4 (Figure S4 in Supporting Information S1). SamudrACE's meridional ocean heat transport shows encouraging agreement to the reference (Figure S20 in Supporting Information S1). In the atmosphere, SamudrACE maintains a stable global energy budget with net energy fluxes and energy budget residuals that track closely with the reference CM4 simulation (Figure S14 in Supporting Information S1). This combination of low bias and stability indicates that SamudrACE may be amenable to additional physical constraints such as energy budget corrections.

3.2. Sea Ice Climatology

SamudrACE predicts a stable and accurate sea ice climatology with a realistic seasonal cycle (Figure 3). The sea ice fraction shown here is derived from SamudrACE's native sea ice concentration described in Section 2.4. SamudrACE skillfully simulates the CM4 seasonal cycle with minimal bias but too little interannual variability in the Southern Hemisphere. In the time mean, SamudrACE tends to have a larger bias in sea ice concentration near the edge of the typical sea ice extent. Notably, the highest positive bias is in the Greenland Sea, corresponding to where SamudrACE also simulates a weaker surface wind stress (not shown). We provide a similar view of sea ice thickness in Figure S13 in Supporting Information S1.

3.3. ENSO

Figure 4 compares ENSO characteristics in CM4 and SamudrACE. The emulator captures fundamental ENSO dynamics, including large-amplitude events, spatial variability patterns, and the characteristic 3-year spectral peak. Sensitivity analysis (Figure S18 in Supporting Information S1) confirms that the full training volume is necessary for Niño 3.4 (Barnston et al., 1997) amplitude to converge toward the reference. In 200-year rollouts initialized from five different years (Figure 4a), SamudrACE exhibits comparable but slightly weaker variability than CM4. Specifically, the emulator shows a bias toward sharper El Niños (max 2.1 K) and weaker La Niñas (min −1.6 K vs. −2.4 K in CM4), and occasionally exhibits decade-long periods of suppressed activity. This asymmetry is further detailed in Figure S7 in Supporting Information S1. Spectral analysis (Figure 4b; see Text S2 in Supporting Information S1 for methods) reveals that while SamudrACE reproduces the 3-year peak, it generally exhibits excessive power in the 2–4 years band and a deficit at lower frequencies (>4 years), indicating that SamudrACE has not fully learned important ocean-atmosphere coupling processes which contribute to quasi-decadal variability (Liu et al., 2022). This high-frequency bias is corroborated by unrealistically high autocorrelation at 36-month lags (Figure S10 in Supporting Information S1). Teleconnections are also well-represented: the regression of precipitation onto the Niño 3.4 index matches the reference (Figure 4c). Comparison with uncoupled ACE2 baselines confirms that ACE2's intrinsic response to SSTs remains largely unchanged by

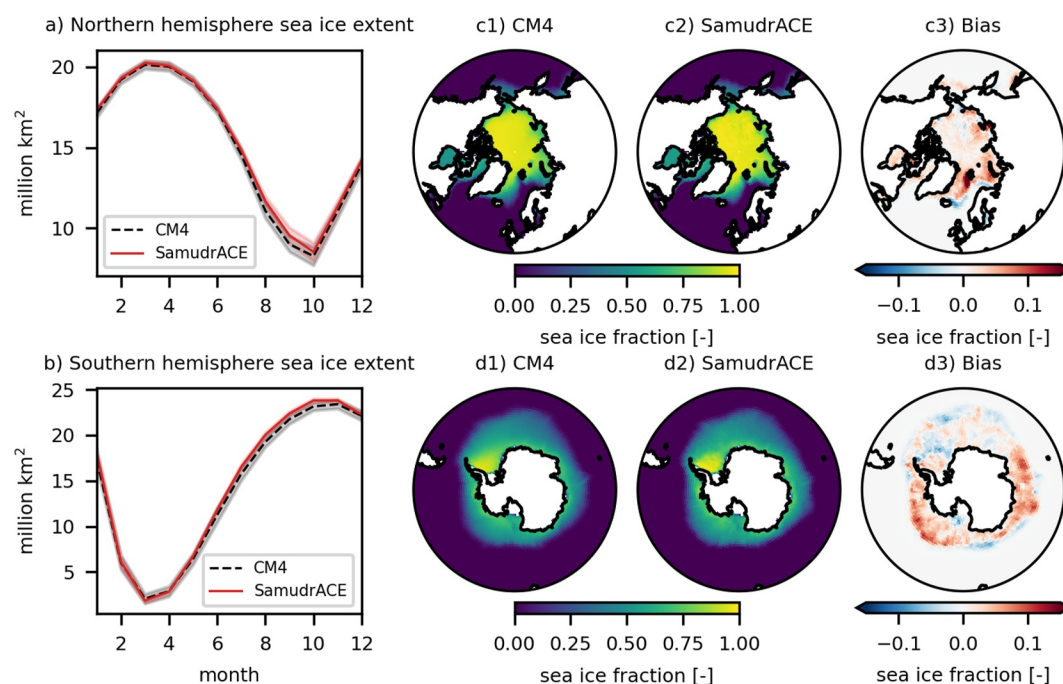


Figure 3. Monthly mean over the 40-year held-out period of (a) Northern and (b) Southern Hemisphere sea ice extent. Shading denotes the interannual standard deviation over 40 years. Panel (c–d) shows the time mean sea ice fraction over the same time period for the CM4 target, SamudrACE, and its bias.

coupled fine-tuning (Figure S19 in Supporting Information S1). Furthermore, despite weaker La Niñas, the vertical structure of equatorial Pacific temperature anomalies remains realistic across ENSO phases (Figures S8 and S9 in Supporting Information S1).

3.4. Decadal Coupled Variability and the Interdecadal Pacific Oscillation

The Interdecadal Pacific Oscillation (IPO) is a pattern of Pacific Ocean variability characterized by multiyear-to-decades-long periods of sustained anomalous SSTs, with important implications for wildlife ecology (Mantua et al., 1997). Like ENSO, the IPO represents emergent coupled variability, though at a much lower periodicity. SamudrACE's generated IPO Tripolar Index (Henley et al., 2015) has a clear pattern of sustained anomalous SST with transitions between extreme polarities on timescales that are similar to those seen in CM4 (Figure S11 in Supporting Information S1). However, consistent with the spectral analysis of ENSO, the magnitude of this low-frequency variability is weaker than in the CM4 reference. While SamudrACE shows clear understanding of atmosphere-ocean coupling to first order, it lacks a more nuanced representation of decadal coupling mechanisms (Capotondi et al., 2023).

4. Conclusions

SamudrACE demonstrates that coupling 3D atmosphere and ocean emulators can produce stable, centuries-long simulations with realistic emergent variability. While promising, our analysis also reveals areas for future improvement. For ENSO and the IPO, the emulator underestimates low-frequency variability on time scales longer than 4 years. We find that increasing the amount of training data may in part mitigate this issue (Text S3 and Figure S18 in Supporting Information S1). Future efforts could also focus on refining the fine-tuning strategy or model architecture to improve the emulator's representation of lower-frequency coupled phenomena, for example, by introducing stochasticity during training. It will also be important to evaluate the effect of the atmosphere-to-ocean coupling timestep and number of rollout steps contributing to the optimization on the emulator fidelity and training cost.

In our approach which couples the pretrained component emulators in physical state space, minor differences in the separate data processing pipelines for uncoupled pretraining of the atmosphere and ocean emulator become

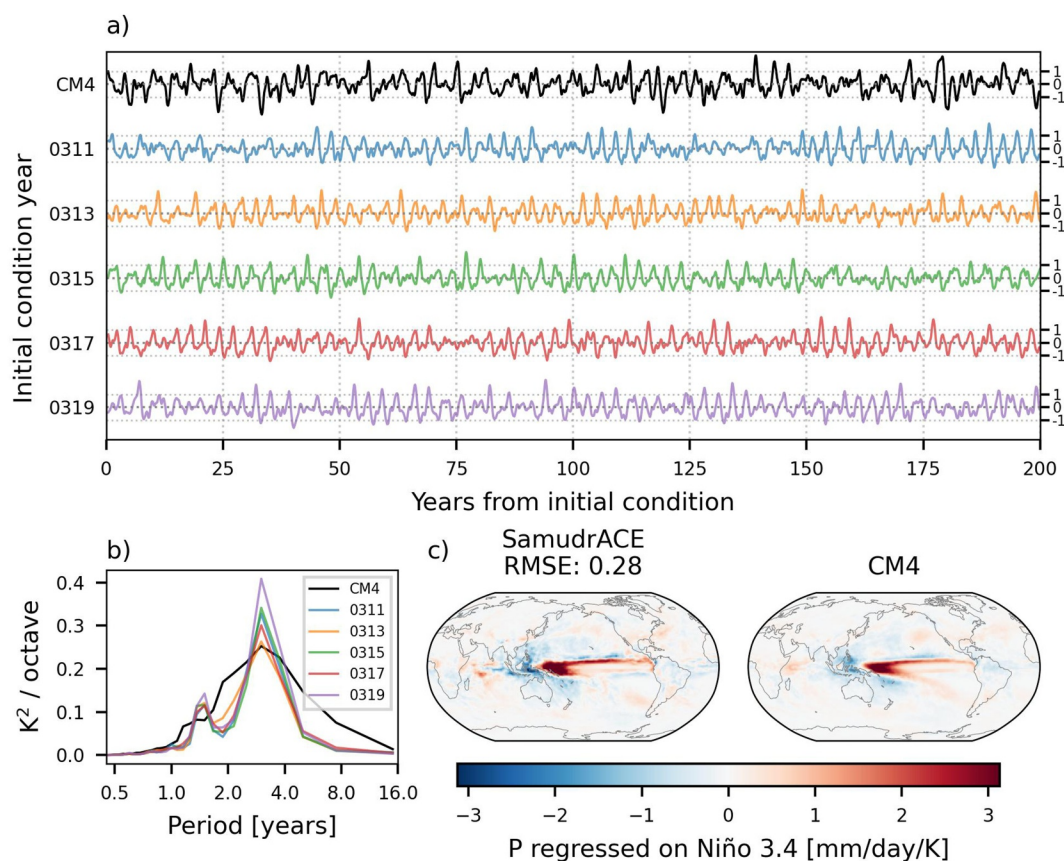


Figure 4. ENSO characteristics in the 200-year CM4 simulation (black) and 5 separate rollouts of SamudrACE starting from different initial conditions (colors): (a) time series of monthly mean Niño 3.4 index; (b) corresponding temporal power spectra (K^2 /octave) computed using Welch's method with 15-year segments (Harris, 2005; Welch, 1967); (c) regression of the spatial pattern of precipitation on the Niño 3.4 index in CM4 and SamudrACE for the held-out 40-year inference period.

readily apparent in the coupled emulator. This is both an advantage and a pitfall of our uncoupled-pretraining-to-coupled-fine-tuning pipeline combined with physics space coupling. Skipping the uncoupled pretraining stage may help to avoid this issue by directly optimizing randomly initialized component emulators in coupled mode, though we found this approach to be computationally inefficient in early experiments. Alternative coupling approaches in component emulator latent space may also be able to circumvent these issues, at the cost of lowered physical interpretability and obfuscation of inconsistencies in uncoupled pretraining. Finally, it may also be possible to avoid offline horizontal regridding of ocean and sea ice outputs entirely by direct emulation of the native-resolution global ocean model outputs and online deterministic or learned regridding when coupling to an atmosphere emulator.

The successful coupled GCM emulation framework of SamudrACE provides a clear pathway toward emulating a complete Earth system by incorporating additional components, such as land and biogeochemical models, opening new avenues for efficient, large-ensemble climate studies. As a natural next step, future work could explore the incorporation of a sea ice emulator able to prognose sea ice concentration at the original 6-hourly temporal resolution used in forcing uncoupled ACE2, and for handling atmosphere-ocean flux and momentum exchange.

In its current configuration and training on a preindustrial control simulation, we do not expect SamudrACE to be capable of generalizing to unseen forced conditions, including abrupt jumps or steady increases in global CO₂ concentrations. However, this study validates the stability and skill of the coupled architecture under control conditions, establishing a foundation for future applications to forced scenarios.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The source code for uncoupled pretraining, coupled fine-tuning, and evaluation of SamudrACE is available in the open source repository <https://github.com/ai2cm/ace> v2026.4.0 (McGibbon et al., 2026). The SamudrACE model weights artifact, initial conditions, and forcing data are available at <https://huggingface.co/allenai/SamudrACE-CM4-piControl>. The full processed 200-year CM4 simulation used for training and testing SamudrACE (approximately 4.1 TB total) can be found in the following public requester-pays bucket in Google Cloud Storage: <gs://ai2cm-public-requester-pays/2026-04-08-SamudrACE-CM4-piControl-GRL>. The source code for figures in this manuscript is available at <https://github.com/ai2cm/SamudrACE-CM4-piControl-GRL> (Duncan, 2026).

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